

Geotopia University



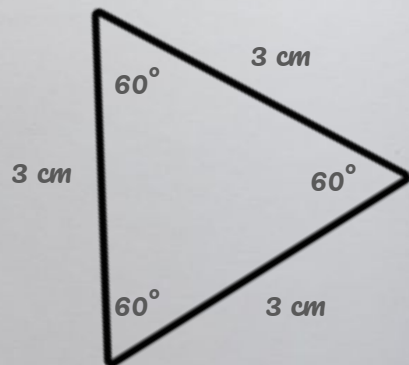
Professor's Note

"Imagination is more important than knowledge. For knowledge is limited, whereas imagination embraces the entire world, stimulating progress, giving birth to evolution."

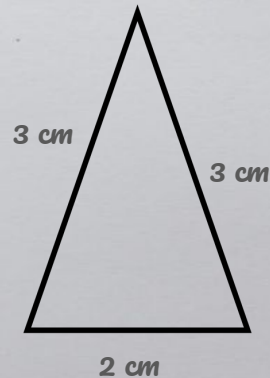
- Albert Einstein

Note #137 Special Triangles

Equilateral Triangle

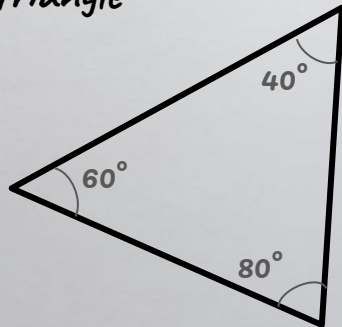


Isosceles Triangle

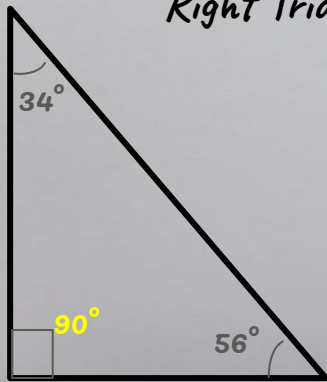


Note #138 3 Types of Triangles

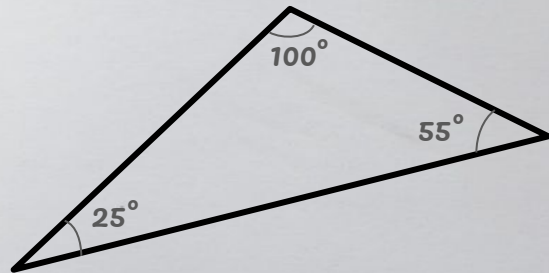
Acute Triangle



Right Triangle



Obtuse Triangle



Facts:

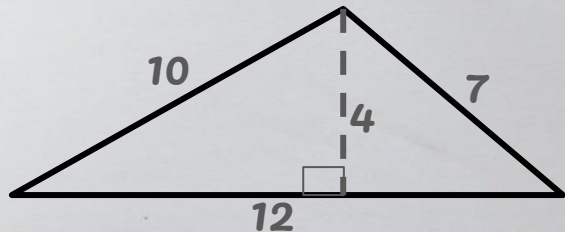
- 1) Angle sum of a triangle is always 180°
- 2) The **base** of a triangle is any of the 3 sides
- 3) The **height** of a triangle is always forming a right angle with the base and ends at one vertex (angle).

Note #139 Area of triangles

The formula for the area of a triangle is $\frac{1}{2}$ times the base times the height.

$$\text{Area} = \frac{1}{2}bh$$

Example:



$$A = \frac{1}{2} \cdot 12 \cdot 4$$

$$A = 6 \cdot 4$$

$$A = 24$$

The area of the triangle is 24 square units.

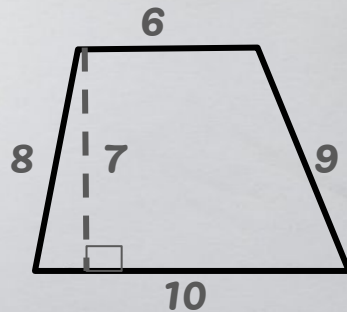
Note #140 Area of Trapezoid

The formula for the area of a trapezoid is $\frac{1}{2}$ the sum of the two bases times the height.

$$A = \frac{1}{2}(b_1 + b_2)h$$

The perimeter is $6+8+10+9=33$

Example:



$$A = \frac{1}{2}(6 + 10)(7)$$

$$A = \frac{1}{2}(16)(7)$$

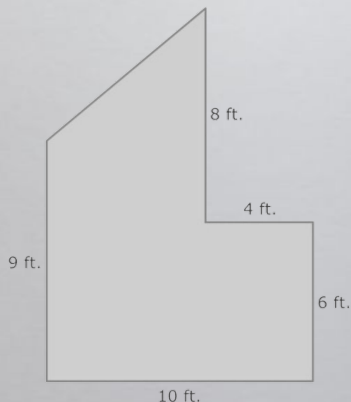
$$A = (8)(7)$$

$$A = 56$$

The area of the triangle is 24 square units.

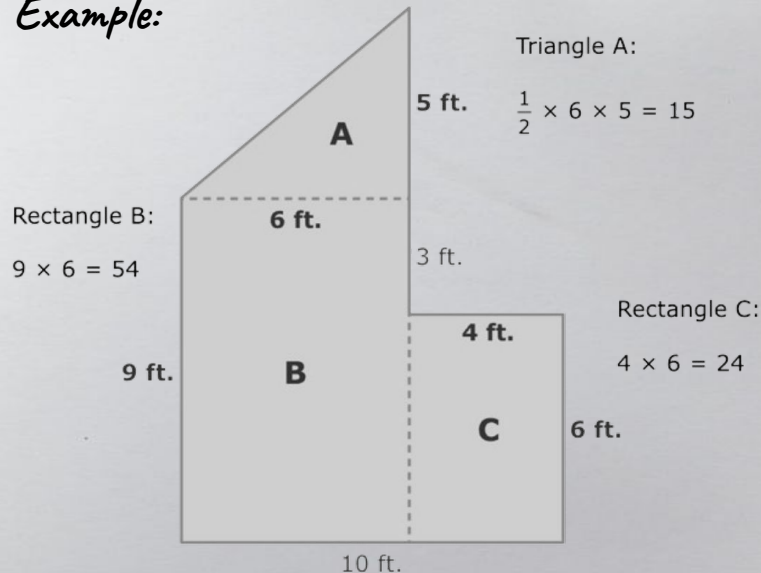
Note #140 Composite Figure

This composite figure (also known as polygon) is made up of basic shapes put together.



To find the area of a composite figure, **break** the composite figure into basic shapes, **find** the area of each basic shape, and then **add** the areas!

Example:



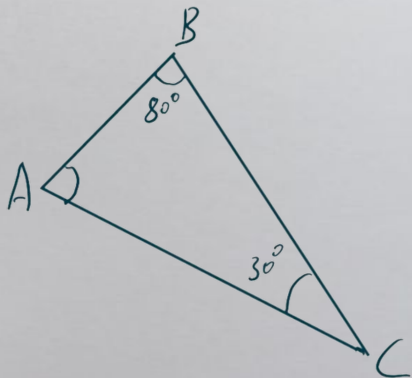
Now, add the areas of the basic shapes.

$$15 + 54 + 24 = 93$$

So, the area of the compound shape is 93 square feet!

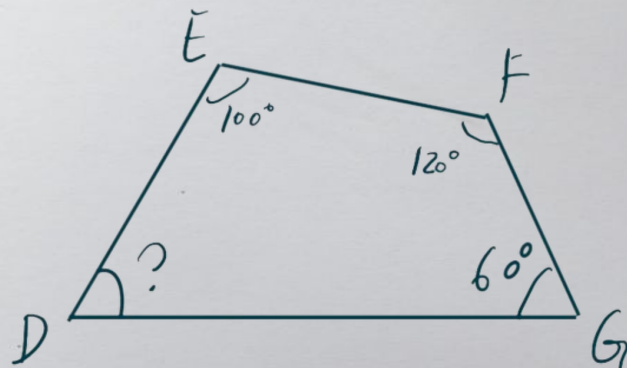
Note #158 Angle Measure

To find the measure of an angle, one way is to use a protractor. However, we can calculate the angle measure as well.



The angle sum of **triangle** is always 180°

$$\text{Angle A} = 180^\circ - 30^\circ - 80^\circ = 70^\circ$$

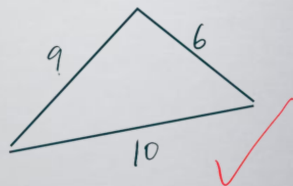


The angle sum of **quadrilateral** is always 360°

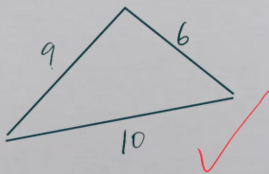
$$\text{Angle D} = 360^\circ - 100^\circ - 120^\circ - 60^\circ = 80^\circ$$

Note #159 Triangle Inequality

Any side of a triangle must be shorter than the **sum** of the other two sides.



$$6 + 9 > 10$$



$$10 - 9 < 6$$

Also, any side of a triangle must be longer than the **difference** between other two sides.

If you have two lengths of 3 and 5, then the possible length for the third side is:

3, 4, 5, 6, 7 (between $5-3$ and $5+3$)

If you have two lengths of 6 and 10, then the possible length for the third side is:

5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
(between $10-6$ and $10+6$)

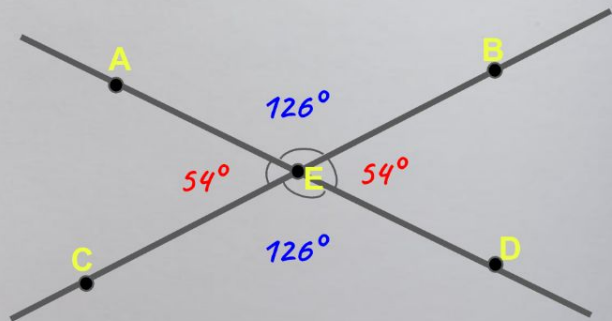
If you have two lengths of 2 and 9, then the possible length for the third side is:

8, 9, 10
(between $9-2$ and $9+2$)

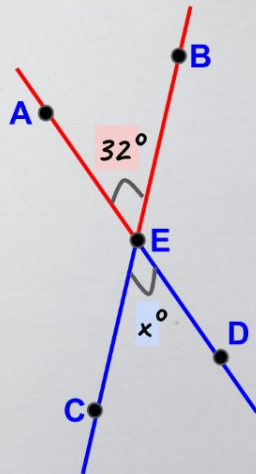
Note #170 Vertical Angles

If two lines are crossing each other, there will be two pairs of **vertical angles**.

Vertical angles are opposite to each others and equal in measure.

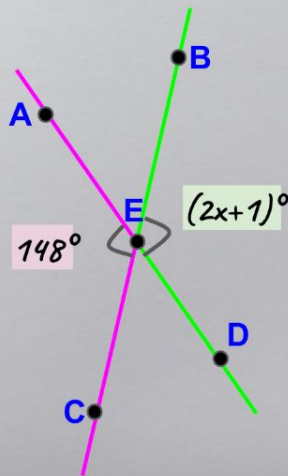


$$\angle AEC = \angle DEB = 126^\circ$$



$$\angle AEB = \angle DEC$$

$$32^\circ = x^\circ$$



$$\angle AEC = \angle DEB$$

$$148^\circ = (2x+1)^\circ$$

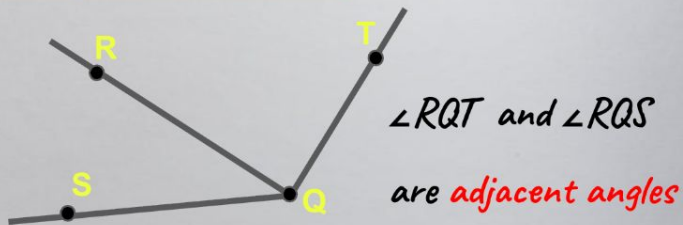
$$148 = 2x+1$$

$$147 = 2x$$

$$73.5 = x$$

Note #171 Supplementary Angles

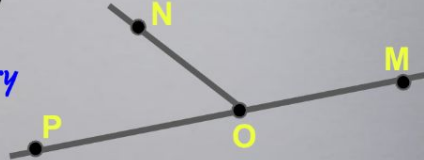
If two angles are side by side, they are **adjacent angles**.



If two angle measures add up to 180° , they are **supplementary angles**.

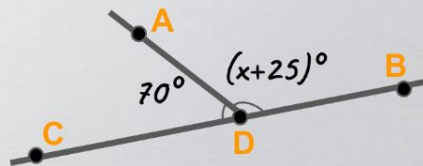
$$\angle NOP + \angle NOM = 180^\circ$$

They are **supplementary**



$\angle ADB$ and $\angle ADC$

are **supplementary**

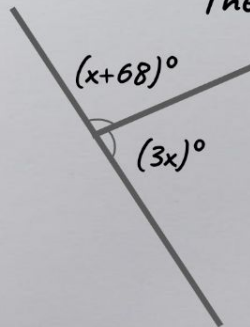


$$70 + x + 25 = 180$$

$$95 + x = 180$$

$$x = 85$$

The two angles are **supplementary**



$$x + 68 + 3x = 180$$

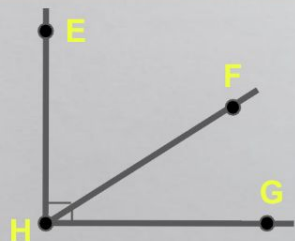
$$68 + 4x = 180$$

$$4x = 112$$

$$x = 28$$

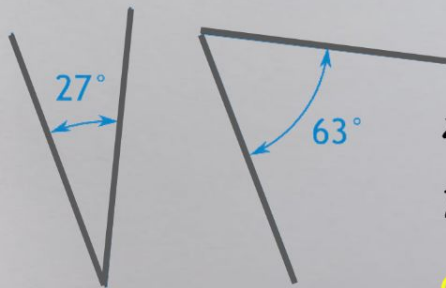
Note #171 Complementary Angles

Two angles are called **complementary** when their measures add up to 90° .



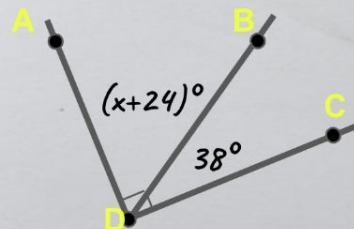
$$\angle EHF + \angle FHG = 90^\circ$$

They are **complementary**



$$27^\circ + 63^\circ = 90^\circ$$

The two angles are **complementary** but **not adjacent**



$\angle ADB$ and $\angle BDC$ are **complementary**

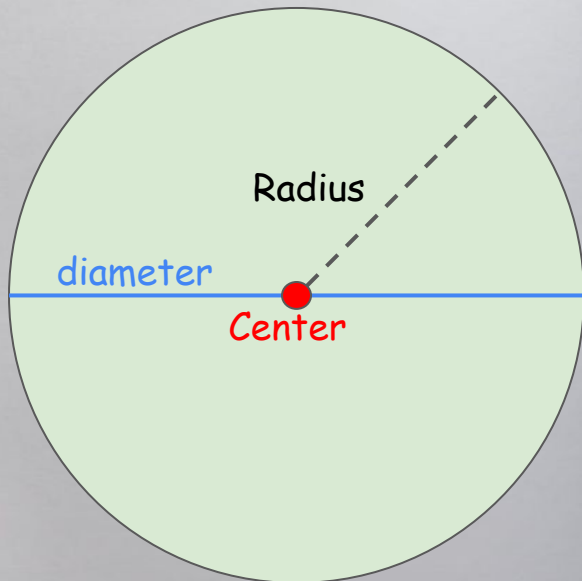
$$x + 24 + 38 = 90$$

$$x + 62 = 90$$

$$x = 28$$

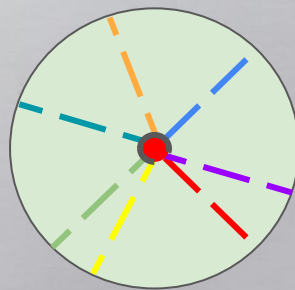
Note #200 Circle

Circle is a curve that is a radius away from the center.

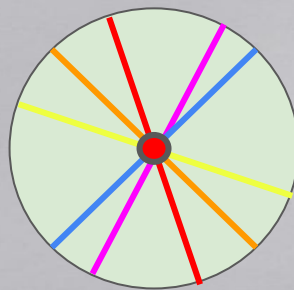


Diameter is always twice the length of radius.

In a circle, you can draw as many radius as you want.

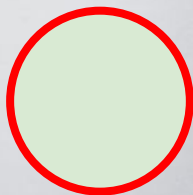


And you can draw as many diameter as you want too.



Note #201 Circumference

The Circumference
is the distance once
around the circle.



Circumference!

$$C = \pi \cdot d = 2 \cdot \pi \cdot r$$

Circumference

Pi

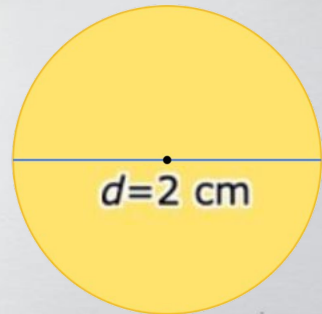
diameter

Double

Pi

Radius

Circumference is 3.14 of
the diameter, or 3.14 of
twice the radius



We know the diameter is 2 cm

$$C = \pi d = \pi \cdot 2 = 2\pi \text{ cm}$$

(in terms of π)

$$2\pi = 2 \cdot 3.14 = 6.28 \text{ cm}$$

(in decimal)



We know the diameter is 2 times
the radius. So diameter is 8 mi.

$$C = 2\pi r = 2 \cdot \pi \cdot 4 = 8\pi \text{ mi}$$

(in terms of π)

$$8\pi = 8 \cdot 3.14 = 25.12 \text{ mi}$$

(in decimal)

Note #202 Area

The area of a circle is π times the radius squared.



Area

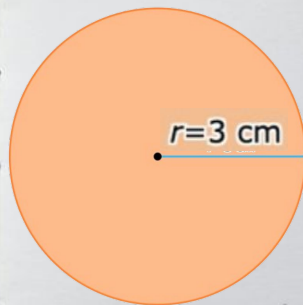
$$A = \pi \cdot r^2$$

Area

Pi

Radius
Square

Area is 3.14 of the radius squares. Multiply radius by itself and then 3.14 for area.



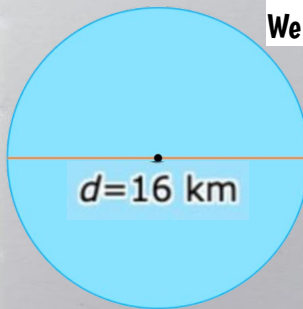
We know the radius is 3 cm.

$$A = \pi r^2 = \pi \cdot 3^2 = 9\pi \text{ mi}^2$$

(in terms of π)

$$9\pi = 9 \cdot 3.14 = 28.26 \text{ mi}^2$$

(in decimal)



Half the diameter is 16 km is the radius 8 km.
We always need radius for area!

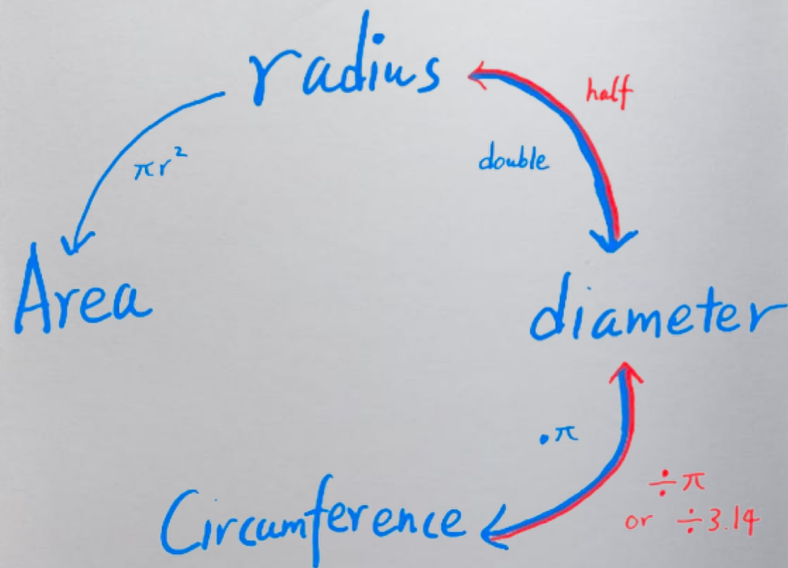
$$A = \pi r^2 = \pi \cdot 8^2 = 64\pi \text{ mi}^2$$

(in terms of π)

$$64\pi = 64 \cdot 3.14 = 200.96 \text{ mi}^2$$

(in decimal)

Note #203 Circumference to Area



If circumference is 15.7 cm,

then diameter is $15.7 \div \pi = 5 \text{ cm}$

then radius is *half* of 5 which is 2.5 cm

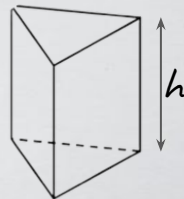
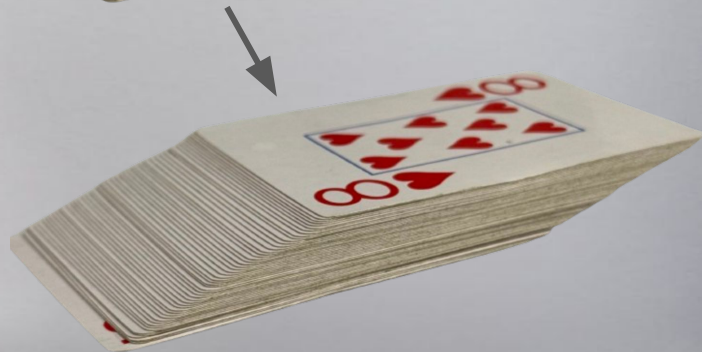
then area is $\pi (2.5^2) = 6.25\pi \text{ cm}^2$
or 19.63 cm^2

Note #225 Volume

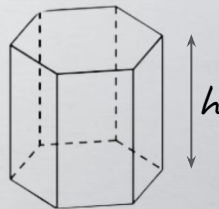
The volume of a prism is calculated by multiplying the area of its base by its height.



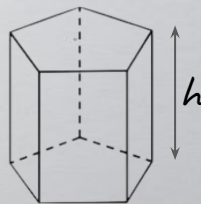
The stack can lean over, but still has the same volume.



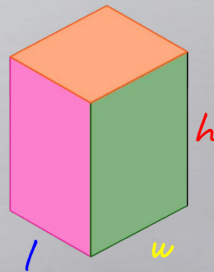
Volume
= base triangle * height



Volume
= base hexagon * height



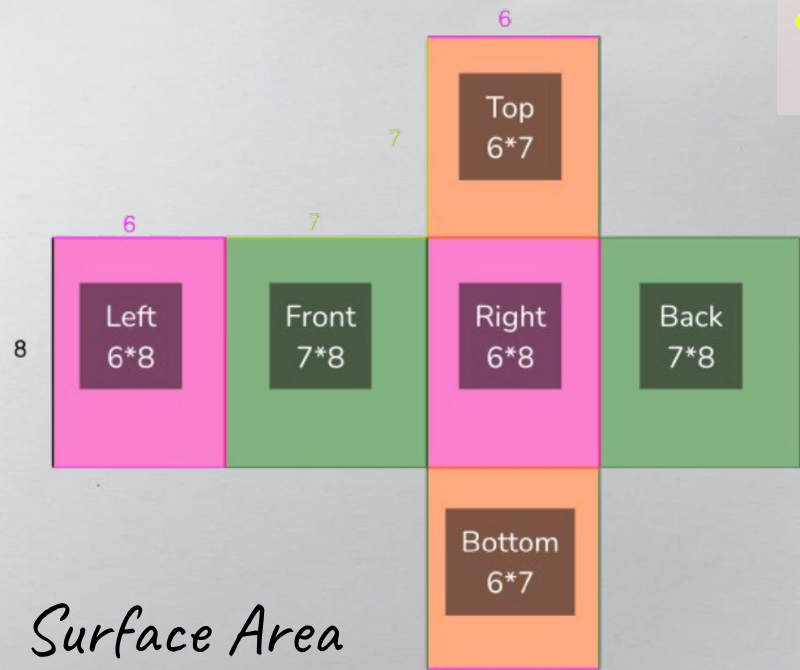
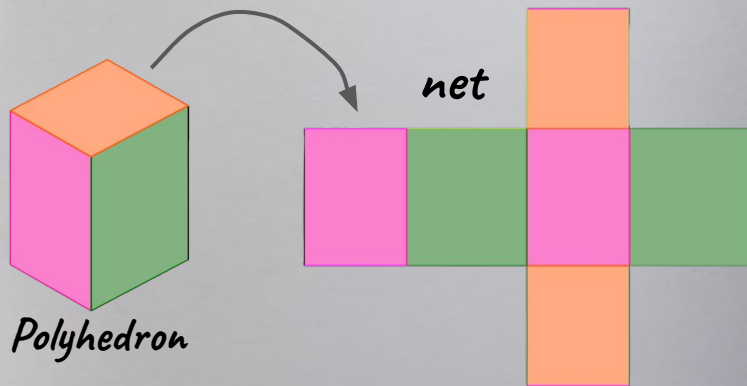
Volume
= base pentagon * height



Volume = base * height
= length * width * height

Note #226 Surface Area of Rectangular Prism (Box)

To find the surface area, we need to "open" the prism. In other words, have the *net* of the prism.



Surface Area

$$= 6 \times 8 + 7 \times 8 + 6 \times 7 + 6 \times 8 + 7 \times 8 + 6 \times 7$$
$$= 292 \text{ unit}^2$$

Or is there a shortcut?

Note #227 Triangular Prism

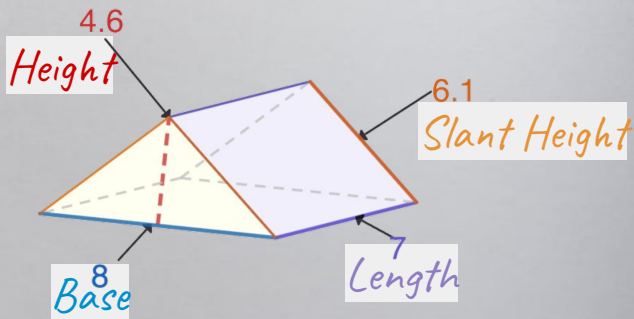
Volume of triangular prism is

$$V = A \cdot l$$

Triangle Area Length

or

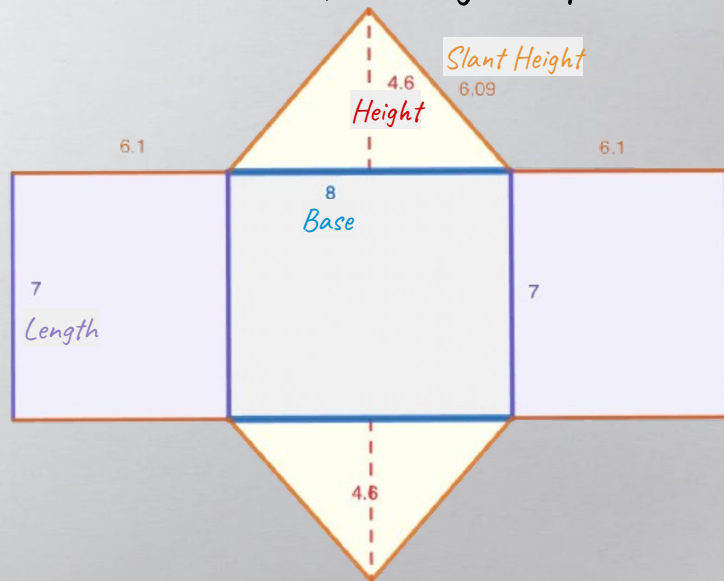
$$V = \left(\frac{1}{2}bh\right) \cdot l$$

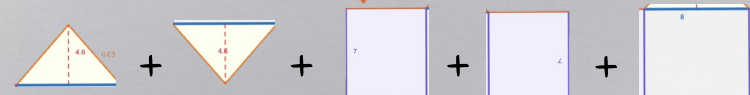


$$V = \left(\frac{1}{2}bh\right) \cdot l = \left(\frac{1}{2} \times 8 \times 4.6\right) \times 7$$
$$= 128.8 \text{ unit}^3$$

(Volume is area of triangle times length)

Surface Area of triangular prism:




$$\frac{1}{2} \times 4.6 \times 8 + \frac{1}{2} \times 4.6 \times 8 + 6.1 \times 7 + 6.1 \times 7 + 8 \times 7$$

$$= 178.2 \text{ unit}^2$$

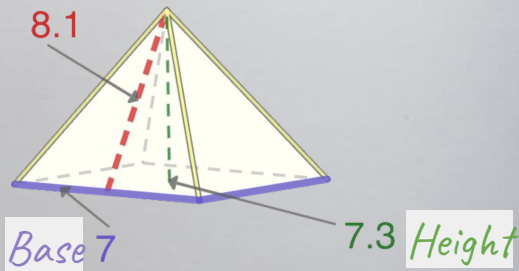
(Surface Area is the sum of 5 faces)

Note #228 Square Pyramid

Volume of a Pyramid is $V = \frac{1}{3} A \cdot h$

Height
Square Area

Slant Height

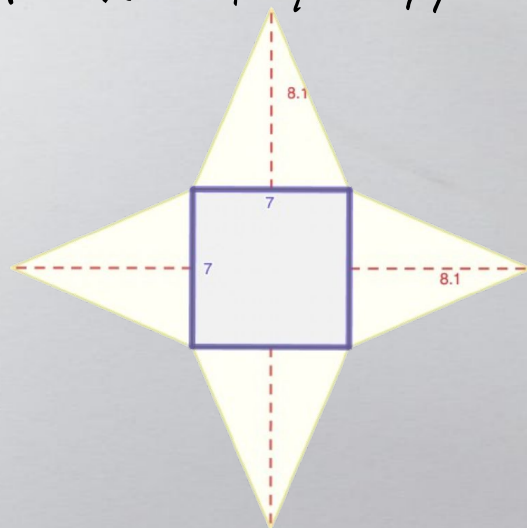


Volume of a square pyramid is
one third the base square times the height.

$$V = \frac{1}{3} A \cdot h = \frac{1}{3} (7 \times 7) \times 7.3$$

$$= 119.23 \text{ unit}^3$$

Surface Area of square pyramid:



A diagram showing the components of the surface area calculation: four triangles and one square. The triangles are arranged in a row, separated by plus signs. The first triangle has a slant height of 8.1. The second triangle has a base of 7 and a slant height of 8.1. The third triangle has a base of 7 and a slant height of 8.1. The fourth triangle has a base of 7 and a slant height of 8.1. The square has a side length of 7.

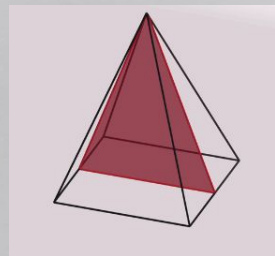
$$4\left(\frac{1}{2} \times 7 \times 8.1\right) + 7 \times 7$$

$$= 162.4 \text{ unit}^2$$

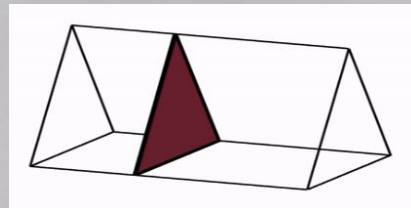
(Surface Area is the sum of 5 faces)

Note #250 Cross Section

A *cross section* is the shape we get when cutting straight through an object.

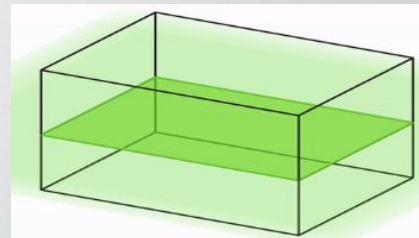


Vertical cut of a square pyramid

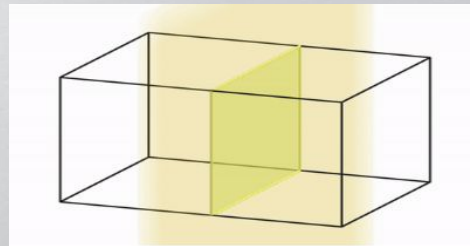


Vertical cut of a triangular prism

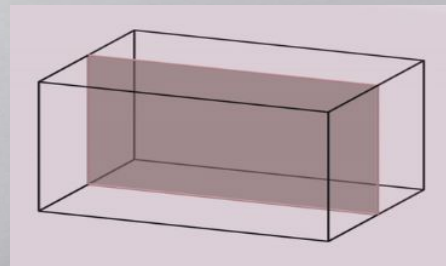
Rectangular Prism



Horizontal cut



Vertical cut from the front



Vertical cut from the right